B8 · Cold drink

Model derivation, step by step

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The model has two parts. The drink is described by one temperature T_d. The bath — ice, brine and undissolved salt — is described by its total enthalpy $\mathcal{H}$, the heat it holds counted from a reference state. One heat flow, $UA(T_d - T_b)$, joins them, and a second, through the styrofoam box, brings heat in from the room. Nothing stirs the bath or the drink.
This document derives the two equations from the conservation of energy and mass and closes them with the phase diagram of salt water and the laws of heat transfer, explaining every step. Physical constants and handbook properties of water, ice, salt, brine and the container materials are put in as numbers, each with a note on what it is (Table 1). Anything that belongs to this rig — masses, sizes, starting temperatures, film coefficients, the box's conductance — stays a symbol, to be measured. Step 13 collects the results into the closed set of equations; Step 14 solves the cases that have a formula.

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How to read this document

What "the model" is. The 24 Oct salted-ice-bath model, `b8_sim.py` in the repository: a lumped drink; a bath described by its enthalpy, with a map of the five states it can be in (liquid, ice on the freezing curve, the eutectic plateau, saturated brine with salt, solid); Bodnar's freezing curve; and the leak of heat through the box. The $\Delta T^{1/4}$ convection law is an option for the heat path. The buffered eutectic bath (COMSOL v1) is a special case.

No stirrer. Nothing stirs the drink or the bath. Fluid moves only by free convection: liquid warmer than its surroundings is lighter and rises, colder liquid sinks. That fixes which convection laws apply (Step 5) and makes the well-mixed bath (assumption A3) the strongest assumption here.

How each step is laid out. It opens with the question it answers and the assumptions it uses. Each line of algebra is followed by what was done and why. It ends in a boxed **Result** and a short note on what the result means.

Which quantities become numbers. A number is put in only for a constant that would be the same in any laboratory: a physical constant, or a handbook property of a pure substance or a standard material. These are listed, and explained, in Table 1. The heat capacities, latent heat, heat of solution and phase diagram are the values `b8_sim.py` uses, so the bath formulas here reproduce its numbers. Everything set by this rig stays a symbol (Table 2), because it has to be measured.

Conventions.

- SI units throughout: J, kg, m, s, W. An enthalpy is in J, a heat flow in W.
- Temperatures T are in °C, because the enthalpy reference and Bodnar's fit are defined at 0 °C. A temperature *difference* is the same in K and °C. Absolute temperature, in K, appears only in the phase-equilibrium derivation (Step 6), written $\mathcal{T}$.
- Salt content w is a mass fraction: kg of NaCl per kg of brine, between 0 and 1. Bodnar's fit uses weight per cent, $W = 100w$.
- A heat flow is positive in the direction its subscript names: $\dot{Q}_{db}$ flows from the drink to the bath, $\dot{Q}_{\text{leak}}$ from the room to the bath.
- Enthalpy reference: liquid water and solid NaCl, both at 0 °C, have zero enthalpy. Any reference would do, because only differences of enthalpy are ever used; this one makes the enthalpy of ice at 0 °C equal to minus its latent heat.

Constants put in as numbers

Each row is a constant that does not depend on this experiment. Where a property changes with temperature, the value is taken where the model uses it, and the note says how much it changes.

Table 1. Constants substituted in this document

Symbol	Value	What it is
Physical constants		
g	$9.81\,\mathrm{m\,s^{-2}}$	Acceleration due to gravity at the Earth's surface. It sets the buoyancy that drives free convection.
R	$8.314\,\mathrm{J\,mol^{-1}K^{-1}}$	Molar gas constant: Boltzmann's constant times Avogadro's number. It converts temperature into energy per mole in the chemical potential.
$\mathcal{T}_0$	$273.15\,\mathrm{K}$	Melting point of pure ice at atmospheric pressure: 0 °C on the absolute scale.
M_w	$0.018015\,\mathrm{kg\,mol^{-1}}$	Molar mass of water, H_2O.
M_s	$0.05844\,\mathrm{kg\,mol^{-1}}$	Molar mass of sodium chloride, NaCl.
ν_i	2	Particles each formula unit of NaCl gives in water: Na^+ and Cl^-.
$j_{0,1}$	2.405	First zero of the Bessel function J_0; it fixes the slowest cooling pattern of a conducting cylinder (Step 3).
Water, ice and salt		
L_f	$333\,550\,\mathrm{J\,kg^{-1}}$	Latent heat of fusion of ice at 0 °C: the heat that melts 1 kg of ice at 0 °C without warming it.
Δh_{fus}	$6009\,\mathrm{J\,mol^{-1}}$	The same per mole, $L_f M_w$.
c_w	$4180\,\mathrm{J\,kg^{-1}K^{-1}}$	Specific heat of liquid water: the heat that warms 1 kg by 1 K (4180–4220 between 30 and 0 °C).
c_i	$2000\,\mathrm{J\,kg^{-1}K^{-1}}$	Specific heat of ice. It falls with temperature, from 2097 at 0 °C to 1950 at −20 °C (IAPWS R10-06); 2000 is its value near −13 °C, where the bath works.
c_h	$864\,\mathrm{J\,kg^{-1}K^{-1}}$	Specific heat of solid NaCl (halite), $50.5\,\mathrm{J\,mol^{-1}K^{-1}}$.
$c_b(w)$	$4180 - 4400\,w$ $\mathrm{J\,kg^{-1}K^{-1}}$	Specific heat of brine. Dissolved salt lowers it; this straight line gives 3300 at $w = 0.20$ and 3160 at w_E, 2–5 % below the handbook values (3370 and 3310, Melinder 2010).
Δh_s	$66\,400\,\mathrm{J\,kg^{-1}}$	Heat absorbed in dissolving 1 kg of NaCl in water ($3.88\,\mathrm{kJ\,mol^{-1}}$, measured at 25 °C in dilute solution). Positive: dissolving salt cools the water.
The NaCl–water phase diagram		
b_1, b_2, b_3	1.78, −0.0442, 5.572×10^{-4}	Coefficients of Bodnar's (1993) fit to the measured freezing curve of salt water, in wt % against degrees below 0 °C.
T_E, w_E	−21.2 °C, 0.2318	The eutectic: the coldest brine that can stay liquid, and its salt content (the end of Bodnar's fit; handbooks give −21.1 °C and 23.3 wt %).
T_p, w_p	+0.1 °C, 0.26285	The peritectic (Swenne 1983): below it the salt that crystallises from saturated brine is hydrohalite, $NaCl{\cdot}2H_2O$, not halite.
K_f	$1.86\,\mathrm{K\,kg\,mol^{-1}}$	Cryoscopic constant of water: freezing-point drop per mole of dissolved particles per kg of water. Derived in Step 6.
Λ_E	$241\,450\,\mathrm{J\,kg^{-1}}$	Heat needed to turn ice and salt at T_E into 1 kg of eutectic brine. Derived in Step 9.
Liquids moving by free convection		
k_d	$0.589\,\mathrm{W\,m^{-1}K^{-1}}$	Thermal conductivity of water at 15 °C, the middle of the drink's 28-to-5 °C range. A soft drink is about 90 % water, so the drink is given water's properties.
ρ_d	$999.1\,\mathrm{kg\,m^{-3}}$	Density of water at 15 °C.
ν	$1.139 \times 10^{-6}\,\mathrm{m^2\,s^{-1}}$	Kinematic viscosity of water at 15 °C: how quickly momentum diffuses through it.
α	$1.408 \times 10^{-7}\,\mathrm{m^2\,s^{-1}}$	Thermal diffusivity of water at 15 °C, $k_d/(\rho_d c_p)$: how quickly heat diffuses through it.
β	$1.51 \times 10^{-4}\,\mathrm{K^{-1}}$	Thermal expansion coefficient of water at 15 °C: the fractional fall in density per kelvin of warming. It is 0 at 4 °C, 0.88×10^{-4} at 10 and 2.07×10^{-4} at 20 °C.
Pr	8.1	Prandtl number of water at 15 °C, ν/α.

Symbol	Value	What it is
k_b	$0.54\,\mathrm{W\,m^{-1}K^{-1}}$	Thermal conductivity of eutectic brine near $-15\,^\circ$C (Melinder 2010).
ρ_b	$1182\,\mathrm{kg\,m^{-3}}$	Density of eutectic brine.
ν_b	$4.6\times10^{-6}\,\mathrm{m^2\,s^{-1}}$	Kinematic viscosity of eutectic brine near $-15\,^\circ$C: four times water's at $15\,^\circ$C, because it is cold and salty.
α_b	$1.38\times10^{-7}\,\mathrm{m^2\,s^{-1}}$	Thermal diffusivity of the same brine. With ν_b it gives a Prandtl number of about 33.
$C_i,\ C_o$	0.62, 0.64	Constants of laminar free convection on a vertical wall, Churchill & Chu (1975), $0.670/[1+(0.492/\mathrm{Pr})^{9/16}]^{4/9}$, at $\mathrm{Pr}=8.1$ (drink) and 33 (brine).
Container and box materials		
c_c	$900\,\mathrm{J\,kg^{-1}K^{-1}}$	Specific heat of aluminium, the can.
k_w	160 (can) or 0.2 (bottle) $\mathrm{W\,m^{-1}K^{-1}}$	Thermal conductivity of the container wall: can-body aluminium alloy (pure aluminium 237), or PET for a plastic bottle.
k_{box}	$0.035\,\mathrm{W\,m^{-1}K^{-1}}$	Thermal conductivity of expanded polystyrene (styrofoam); 0.033–0.038 depending on the foam's density.

Symbols kept as symbols

Source column: **M** measured on this rig, **C** chosen by the experimenter, **D** derived here, **S** a state variable that changes during a run.

Table 2. Symbols that stay symbols

Symbol	Meaning	Unit	Src
T_d	drink temperature (its well-mixed core)	°C	S
T_b	bath temperature	°C	D (from $\mathcal{H}$)
T_{env}	room air outside the box	°C	M
θ	drink–bath difference, $T_d - T_b$	K	D
$m_d,\ c_d$	drink mass; drink specific heat (about 4000 for a sugary drink, 4180 for water or a diet drink)	kg, $\mathrm{J\,kg^{-1}K^{-1}}$	M
m_c	container mass	kg	M
C_d	heat capacity of drink + container	$\mathrm{J\,K^{-1}}$	D
$r_i,\ r_o,\ H$	container inner radius, outer radius, height of the wetted wall	m	M
t_w	wall thickness	m	M
$A_i,\ A_o$	inner and outer wetted areas	$\mathrm{m^2}$	M
$h_i,\ h_o$	drink-side and bath-side film coefficients	$\mathrm{W\,m^{-2}K^{-1}}$	D/M
R_i, R_w, R_o, R_c	inner film, wall, outer film and contact (label) resistances	$\mathrm{K\,W^{-1}}$	D
$U,\ UA$	overall heat-transfer coefficient; conductance	$\mathrm{W\,m^{-2}K^{-1}}$, $\mathrm{W\,K^{-1}}$	M (fit)
Bi, Nu, Ra	Biot, Nusselt and Rayleigh numbers	—	D
$\mathcal{H}$	total enthalpy of the bath	J	S
C_t	heat capacity of the tank glass, if counted with the bath	$\mathrm{J\,K^{-1}}$	M
$m_W,\ m_S$	all the water (ice + liquid), all the NaCl	kg	M
$m_i,\ m_l,\ m_h$	ice, brine, undissolved salt	kg	S
m_m	mass of ice melted so far	kg	S
$w,\ \bar{w}$	salt fraction of the brine; overall salt fraction $m_S/(m_W+m_S)$	—	S, D
$T_{\mathrm{liq}}(w)$	freezing temperature of brine of salt fraction w	°C	D
$w_{\mathrm{sat}}(T)$	salt fraction of saturated brine at T	—	D
ϑ	degrees below zero on the freezing curve, $-T_{\mathrm{liq}}$	K	D

Symbol	Meaning	Unit	Src
$L_{\text{eff}},\ C_{\text{eff}}$	effective latent heat; effective heat capacity of the bath	$\mathrm{J\,kg^{-1}}$, $\mathrm{J\,K^{-1}}$	D
C_B	heat capacity of the ice and brine together	$\mathrm{J\,K^{-1}}$	D
UA_{box}	conductance of the box from room to bath	$\mathrm{W\,K^{-1}}$	M
$A_{\text{box}},\ \delta_{\text{box}}$	foam area and thickness	$\mathrm{m^2}$, m	M
$\Delta\rho$	density difference driving the brine film	$\mathrm{kg\,m^{-3}}$	D
$a_w,\ \varphi,\ b$	water activity, osmotic coefficient, molality	—, —, $\mathrm{mol\,kg^{-1}}$	D
$\tau,\ \tau_0,\ t^*,\ \tau_c$	time constants	s	D
$n,\ a$	exponent and prefactor in $\dot{Q}_{db} = a\theta^n$	—, $\mathrm{W\,K^{-n}}$	D/M
$T^*,\ T_{fd}$	target drink temperature; freezing point of the drink	°C	C, M
m_{buf}	salt to keep undissolved as a buffer	kg	C

Step 0 — System, control volumes and assumptions

The question. What are the parts of the rig, which of them store heat, and along which paths does heat move between them?

A *control volume* is a region with a boundary drawn round it, kept account of: what crosses the boundary changes what is inside. The rig splits into two volumes that store heat — the drink and the bath — and two that only resist its flow — the container wall and the box. Everything outside the box is the room, so large that the heat it gives up does not change its temperature T_{env}.

Control volume	Contents	State	Exchanges heat with
Drink (D)	liquid + container shell	T_d	the bath, through the wall
Bath (B)	ice, brine, undissolved NaCl	$\mathcal{H}$	the drink; the room, through the box
Container wall	aluminium or PET shell, label	—	resistance only
Box	EPS walls + lid	—	resistance only

Thermal circuit. Heat flows like current in an electrical circuit: temperatures play the part of voltages, heat flows of currents, thermal resistances ($\mathrm{K\,W^{-1}}$) of electrical ones, and heat capacities of capacitors, which store heat as their temperature changes.

$$\underbrace{T_d}_{C_d} \xrightarrow{R_i} T_{wi} \xrightarrow{R_c+R_w} T_{wo} \xrightarrow{R_o} \underbrace{T_b}_{\mathcal{H}} \xleftarrow{1/UA_{\text{box}}} T_{\text{env}}$$

Heat leaves the drink through the film of drink on the inside of the wall (R_i), crosses the label and the wall ($R_c + R_w$), and enters the bath through the film of brine on the outside (R_o). The wall's two faces are at T_{wi} and T_{wo}. From the other side, heat from the room leaks through the box into the bath. The drink stores heat in C_d; the bath stores it in its enthalpy $\mathcal{H}$, much of it as the latent heat of its ice.

Assumptions, each with the reason it is made and where it is tested.

A1 Drink well-mixed. One temperature T_d describes the liquid core; all the change of temperature sits in thin boundary layers at the wall. Free convection keeps the core mixed; Step 3 tests when that is enough.

A2 Wall and tank store no heat, or their heat capacity is added to the system they touch: the can's to C_d, the tank glass's (C_t) to the bath. A can wall 0.1 mm thick holds about 1 % as much heat per kelvin as its drink.

A3 Bath well-mixed and at phase equilibrium at every instant. One temperature T_b and one salt fraction w describe the whole bath, and ice, brine and salt are always in equilibrium. With no stirrer this is the strongest assumption: brine next to the can warms first. The report and the browser simulator relax it, splitting the bath into rings round the

can and melting and dissolving at finite rates.

A4 Quasi-steady resistances. The thin films and the wall reach their steady temperature profile much faster (seconds) than T_d or T_b change (minutes), so each is described by a fixed resistance.

A5 Constant pressure, lid on. Nothing evaporates or condenses, and no frost forms. Radiation inside the box is counted in UA_{box}.

A6 Constant properties. Each specific heat, latent heat and heat of solution is one number; the brine's specific heat depends on its salt content through one fixed straight line. The exceptions are h_i and β in the convection laws (Step 5).

A7 The stable phase diagram, with salt booked as halite. The freezing curve, the solubility and the eutectic are the measured equilibrium ones. Undissolved salt is counted as anhydrous NaCl (halite) in the mass and energy balances. Below +0.1 °C the stable salt solid is really hydrohalite, $\text{NaCl}{\cdot}2\text{H}_2\text{O}$, which locks up $2M_w/M_s = 0.62\,\text{kg}$ of water per kg of salt; that bound water and its heat are left out. Nothing supercools and no metastable phase forms.

A8 Drink fully immersed to a marked waterline, so its wetted area is fixed.

A9 Ideal sensors. Probe lag and offsets are corrections to the measurements, not terms in the model.

Step 1 — First law for the drink

The question. How fast does the drink's temperature fall for a given heat flow out of it?

1.1 The first law at constant pressure. The drink and its can form a closed system: no mass crosses its boundary. For a closed system the first law of thermodynamics says that its internal energy E changes by the heat that enters it minus the work it does, $\mathrm{d}E = \delta Q - p\,\mathrm{d}V$. At constant pressure (A5) the work of expansion is folded in by using the enthalpy $H = E + pV$, since $\mathrm{d}H = \mathrm{d}E + p\,\mathrm{d}V$ when p is fixed. So $\mathrm{d}H = \delta Q$: at constant pressure, *heat in equals the rise of enthalpy.* The only heat crossing the boundary is $\dot{Q}_{db}$, flowing out, so

$$\frac{\mathrm{d}H_D}{\mathrm{d}t} = -\dot{Q}_{db}.$$

1.2 The enthalpy of the drink. The specific heat c is defined by $\mathrm{d}h = c\,\mathrm{d}T$ for 1 kg. Each kilogram of drink holds $c_d(T_d - T_{\text{ref}})$ above a reference temperature, and each kilogram of can $c_c(T_d - T_{\text{ref}})$, since the can is at the drink's temperature (A1, A2). Adding the two masses, with constant specific heats (A6),

$$H_D = (m_d c_d + m_c c_c)(T_d - T_{\text{ref}}) \equiv C_d\,(T_d - T_{\text{ref}}), \qquad \frac{\mathrm{d}H_D}{\mathrm{d}t} = C_d \frac{\mathrm{d}T_d}{\mathrm{d}t}.$$

The reference drops out on differentiating. With aluminium's specific heat, $c_c = 900\,\text{J}\,\text{kg}^{-1}\text{K}^{-1}$:

$$C_d = m_d c_d + 900\,m_c \qquad (\text{J}\,\text{K}^{-1}).$$

The drink's own specific heat c_d stays a symbol because it depends on the drink: sugar lowers it, from water's 4180 to about $4000\,\text{J}\,\text{kg}^{-1}\text{K}^{-1}$ for a sugary soft drink.

Result 1 — drink energy balance

$$C_d \frac{\mathrm{d}T_d}{\mathrm{d}t} = -\dot{Q}_{db}, \qquad C_d = m_d c_d + 900\,m_c$$

What it means. C_d is the heat the drink and can give up for each kelvin they cool, so the drink cools at $\dot{Q}_{db}/C_d$ kelvin per second. A can's 13 g of aluminium adds $12\,\mathrm{J\,K^{-1}}$ to the drink's 1300 or so: about 1 %. Weigh the drink full and empty to get m_d; the outside volume $\pi r_o^2 H$ would include the gas in the headspace.

Step 2 — Heat path drink → bath as series resistances

The question. How does the heat flow out of the drink depend on the two temperatures, and on the films and the wall it crosses?

2.1 Each layer carries the same heat flow. By A4 the films and the wall store no heat, so whatever heat enters one face of a layer leaves the other. The same $\dot{Q}_{db}$ therefore crosses all three, each driven by the temperature drop across it and opposed by its resistance:

$$\dot{Q}_{db} = \frac{T_d - T_{wi}}{R_i} = \frac{T_{wi} - T_{wo}}{R_w} = \frac{T_{wo} - T_b}{R_o}.$$

2.2 Film resistances. Next to a wall, a liquid moving by convection is held still in a thin film, which heat crosses only by conduction. Newton's law of cooling, $\dot{Q} = hA\,\Delta T$, sums that up in a film coefficient h, the heat flow per unit area per kelvin of drop. Comparing with $\dot{Q} = \Delta T/R$,

$$R_i = \frac{1}{h_i A_i}, \qquad R_o = \frac{1}{h_o A_o}.$$

The film coefficients depend on how the liquid moves; Step 5 derives them.

2.3 Wall resistance of a cylinder. Fourier's law says heat flows down a temperature gradient in proportion to it: the flux is $-k_w\,\mathrm{d}T/\mathrm{d}r$ through a solid of conductivity k_w. Through the side wall of a cylinder of height H, it crosses the area $2\pi r H$ at radius r, and the total $\dot{Q}$ is the same at every radius (no storage, A4):

$$\dot{Q} = -k_w(2\pi r H)\frac{\mathrm{d}T}{\mathrm{d}r} \;\Rightarrow\; \int_{r_i}^{r_o} \frac{\mathrm{d}r}{r} = -\frac{2\pi k_w H}{\dot{Q}} \int_{T_{wi}}^{T_{wo}} \mathrm{d}T$$

$$\Rightarrow\; \ln\frac{r_o}{r_i} = \frac{2\pi k_w H}{\dot{Q}}\,(T_{wi} - T_{wo}) \;\Rightarrow\; R_w = \frac{T_{wi} - T_{wo}}{\dot{Q}} = \frac{\ln(r_o/r_i)}{2\pi k_w H}.$$

The variables were separated (r on one side, T on the other) and both sides integrated across the wall. For a thin wall, $t_w = r_o - r_i \ll r_i$, the logarithm is $\ln(1 + t_w/r_i) \approx t_w/r_i$, so $R_w \approx t_w/(k_w A)$ with $A = 2\pi r_i H$: the wall acts as a flat slab. A label sleeve adds a contact resistance $R_c = R_c''/A$, with R_c'' its resistance per unit area, in series.

2.4 Add the series. Rearrange each equality in 2.1 as a temperature drop, $T_d - T_{wi} = \dot{Q}_{db} R_i$ and so on, including the label's, and add them. The wall temperatures cancel in pairs:

$$T_d - T_b = \dot{Q}_{db}\,(R_i + R_c + R_w + R_o).$$

Resistances in series add, because their temperature drops add while the heat flow is common.

2.5 Faces in parallel. The side, the base and (if submerged) the top of the container are separate paths j side by side, each with its own films and wall. Each carries its own heat flow driven by the

same $T_d - T_b$, so the flows add, and with them the conductances (inverse resistances):

$$UA = \sum_j \left(\frac{1}{h_{i,j} A_{i,j}} + R_{c,j} + R_{w,j} + \frac{1}{h_{o,j} A_{o,j}} \right)^{-1},$$
$$\frac{1}{U} = \frac{1}{h_i} + R_c'' + \frac{t_w}{k_w} + \frac{1}{h_o} \quad \text{(thin wall, } A_i \approx A_o \approx A\text{)}.$$

The second line, for one face with a thin wall, multiplies every resistance by the common area: U is the overall coefficient per unit area, and UA the conductance in $\mathrm{W\,K^{-1}}$.

2.6 What a probe on the wall measures. Dividing two of the drops in 2.4 gives the share of the temperature difference each part takes, as in a potential divider:

$$\frac{T_d - T_{wo}}{T_{wo} - T_b} = \frac{R_i + R_c + R_w}{R_o}.$$

A probe taped to the outside of the can therefore reads neither the drink nor the bath.

Can versus bottle. With can-body aluminium, $k_w = 160\,\mathrm{W\,m^{-1}K^{-1}}$, a wall 0.1 mm thick has $t_w/k_w = 6 \times 10^{-7}\,\mathrm{m^2K\,W^{-1}}$; a film with $h = 300$ has $1/h = 3 \times 10^{-3}$, five thousand times more. So R_w drops out for a can. With PET, $k_w = 0.2$, a wall 0.3 mm thick has $t_w/k_w = 1.5 \times 10^{-3}$, half a film's resistance: a bottle's wall matters. Compare containers by U, not by their cooling time, which also depends on how much drink they hold (Step 4).

Result 2 — heat flow from drink to bath

$$\dot{Q}_{db} = UA\,(T_d - T_b), \qquad \frac{1}{UA} = R_i + R_c + R_w + R_o$$

What it means. The whole path from drink to bath is one conductance UA, and the heat flow is proportional to the temperature difference — as long as the film coefficients do not themselves depend on it, which Step 5 revisits.

Step 3 — Biot number and when lumping is allowed

The question. Is it fair to describe the whole drink by one temperature (A1)?

3.1 The usual test. Heat leaving a body meets two resistances in series: inside it, across a length L_c of material of conductivity k, about L_c/k per unit area; outside it, $1/U_{\mathrm{ext}}$. Their ratio is the Biot number,

$$\mathrm{Bi} = \frac{U_{\mathrm{ext}} L_c}{k_d} = 1.70\,U_{\mathrm{ext}} L_c,$$
$$\frac{1}{U_{\mathrm{ext}}} = R_c'' + \frac{t_w}{k_w} + \frac{1}{h_o}, \qquad L_c = \frac{V}{A} \text{ or } r_i,$$

with water's conductivity $k_d = 0.589\,\mathrm{W\,m^{-1}K^{-1}}$ (U_{ext} in $\mathrm{W\,m^{-2}K^{-1}}$, L_c in m). If $\mathrm{Bi} < 0.1$, the inside resistance is small, the temperature inside hardly varies, and a body that conducts can be lumped. A 330 ml can has $V/A \approx 1.3$ cm; with U_{ext} of a few hundred $\mathrm{W\,m^{-2}K^{-1}}$, Bi is about 5 to 10. A still drink fails the test. The tiny Biot number often quoted for a can uses the aluminium's k_w and t_w: it belongs to the wall, not the drink.

3.2 How slowly a still drink would cool. If the drink only conducted, the heat equation $\partial T/\partial t = \alpha \nabla^2 T$ would hold inside it. In a long cylinder whose wall is held at a fixed temperature, separating variables gives patterns $J_0(\lambda r)\, e^{-\alpha \lambda^2 t}$, where J_0 is a Bessel function; the wall condition

$J_0(\lambda r_i) = 0$ allows $\lambda = j_{0,1}/r_i$ for the slowest pattern, $j_{0,1} = 2.405$. That pattern decays with time constant

$$\tau_{\text{cond}} = \frac{r_i^2}{\alpha\, j_{0,1}^2} = \frac{r_i^2}{1.408 \times 10^{-7} \times 5.783} = 1.23 \times 10^6\, r_i^2 \ \text{ s} \qquad (r_i \text{ in m}),$$

22 minutes for a can of radius 3.3 cm: conduction alone would take more than an hour to cool it.

3.3 The rescue: free convection. The drink is not still. Liquid cooled at the wall is denser and sinks in a thin layer down the wall, while the core is stirred by the circulation this sets up. Step 5 shows that the layer is $\delta \sim H\,\mathrm{Ra}_H^{-1/4}$ thick, where Ra_H is the Rayleigh number; with water's properties,

$$\delta \sim \frac{H^{1/4}}{310\,\Delta T_i^{1/4}} = 3.2 \times 10^{-3}\, H^{1/4}\, \Delta T_i^{-1/4} \ \text{ m},$$

with H in m and ΔT_i in K: about 1 mm for a can 11.7 cm tall and a 10 K drop: far thinner than the radius. All the temperature change sits in that layer, whose resistance is R_i, already in the circuit. The measured speed-up over conduction is $E = \tau_{\text{cond}}/\tau_{\text{measured}}$.

Result 3 — lumping criterion

A1 holds if $\delta \sim H\,\mathrm{Ra}_H^{-1/4} \ll r_i$ and $|T_{\text{top}} - T_{\text{bottom}}| \ll T_d - T_b$ (not $\mathrm{Bi} < 0.1$)

What it means. The drink is a mixed core inside a thin cold layer, not a conducting solid. The test is whether the layer is thin and the core uniform, which a probe at the top and one at the bottom of the drink can check: cold liquid collects at the bottom, so the core does stratify slowly.

Step 4 — Fixed-bath solution

The question. If the bath stays at one temperature, how does the drink cool?

4.1 Combine Results 1 and 2. Put $\dot{Q}_{db} = UA(T_d - T_b)$ into $C_d\,\mathrm{d}T_d/\mathrm{d}t = -\dot{Q}_{db}$. With T_b constant, $\mathrm{d}\theta/\mathrm{d}t = \mathrm{d}T_d/\mathrm{d}t$ for the difference $\theta = T_d - T_b$, so

$$\frac{\mathrm{d}\theta}{\mathrm{d}t} = -\frac{\theta}{\tau}, \qquad \tau \equiv \frac{C_d}{UA}.$$

The difference falls at a rate proportional to itself; τ (seconds) is the drink's heat capacity over its conductance.

4.2 Separate and integrate. Put everything in θ on the left and in t on the right, and integrate from the start, where $\theta = \theta_0$:

$$\int_{\theta_0}^{\theta} \frac{\mathrm{d}\theta'}{\theta'} = -\int_0^t \frac{\mathrm{d}t'}{\tau} \ \Rightarrow\ \ln\frac{\theta}{\theta_0} = -\frac{t}{\tau}.$$

4.3 Solve for the temperature. Exponentiate both sides and add back T_b: this gives Result 4. In each time τ the remaining difference shrinks by the factor $e = 2.718$.

4.4 Derived quantities. Setting $T_d = T^*$ and solving for t gives the time to reach a target; differentiating at $t = 0$ gives the initial cooling rate:

$$t_{T^*} = \tau \ln\frac{T_{d0} - T_b}{T^* - T_b}, \qquad \left.\frac{\mathrm{d}T_d}{\mathrm{d}t}\right|_0 = -\frac{T_{d0} - T_b}{\tau}.$$

Two baths at T_{b1} and T_{b2} can be compared in two ways:

$$\frac{t_{T^*}(T_{b1})}{t_{T^*}(T_{b2})} = \frac{\ln[(T_{d0} - T_{b1})/(T^* - T_{b1})]}{\ln[(T_{d0} - T_{b2})/(T^* - T_{b2})]}, \qquad \frac{\dot{T}_d(0)|_{T_{b1}}}{\dot{T}_d(0)|_{T_{b2}}} = \frac{T_{d0} - T_{b1}}{T_{d0} - T_{b2}}.$$

The two ratios differ: "how much faster does the colder bath cool the drink" has two honest answers, one for the time to a target and one for the initial rate.

4.5 Geometry inside τ. The drink's heat capacity is about $\rho_d c_d V$ for a volume V, so $\tau = (\rho_d c_d / U)\,(V/A)$: at the same U, a container with more drink per unit of wetted area cools more slowly.

Result 4 — fixed-bath cooling curve

$$T_d(t) = T_b + (T_{d0} - T_b)\, e^{-t/\tau}, \qquad \tau = \frac{C_d}{UA}$$

What it means. A plot of $\ln(T_d - T_b)$ against time is a straight line of slope $-1/\tau$; from its slope and the weighed C_d, $UA = C_d/\tau$.

Step 5 — Convection closures and the power-law cooling curve

The question. What sets the film coefficients when nothing is stirred, and how does that change the cooling curve?

5.1 Drink side: laminar free convection. In the layer next to the wall the drink is colder by about $\Delta T_i = T_d - T_{wi}$, so denser by $\rho\beta\,\Delta T_i$: a buoyancy force $\rho g\beta\,\Delta T_i$ per unit volume drives it down the wall. In a layer of thickness δ moving at speed u, viscosity resists with a force $\rho\nu u/\delta^2$; balancing the two gives $u \sim g\beta\Delta T_i\delta^2/\nu$. Heat carried along the wall over its height H, $u\Delta T_i/H$, must balance heat conducted across the layer, $\alpha\Delta T_i/\delta^2$. Eliminating u gives $\delta \sim H\,\mathrm{Ra}_H^{-1/4}$, and since the film coefficient is about k_d/δ,

$$\mathrm{Ra}_H = \frac{g\beta\,\Delta T_i H^3}{\nu\alpha}, \qquad \mathrm{Nu}_H = \frac{h_i H}{k_d} = C_i\,\mathrm{Ra}_H^{1/4}.$$

The Rayleigh number compares buoyancy with the two diffusions, of momentum and of heat, that resist it; the Nusselt number compares the film coefficient with conduction over the height. The scaling fixes the power $\frac{1}{4}$; the constant comes from experiments on vertical walls, $C_i = 0.62$ for water (Table 1). With water at 15 °C,

$$\mathrm{Ra}_H = \frac{9.81 \times 1.51 \times 10^{-4}}{1.139 \times 10^{-6} \times 1.408 \times 10^{-7}}\,\Delta T_i H^3 = 9.22 \times 10^9\,\Delta T_i H^3.$$

Solving the Nusselt relation for h_i and putting in the numbers:

$$h_i = C_i k_d \left(\frac{g\beta}{\nu\alpha H}\right)^{1/4} \Delta T_i^{1/4} = 0.62 \times 0.589 \times 310\; H^{-1/4}\,\Delta T_i^{1/4} \equiv K_i\,\Delta T_i^{1/4},$$

$$K_i = 112\, H^{-1/4}$$

in $\mathrm{W\,m^{-2}K^{-5/4}}$ with H in m. For an upright can 11.66 cm tall, $K_i = 192$, and at a 25 K drop $h_i = 430\,\mathrm{W\,m^{-2}K^{-1}}$, close to the 400 measured on a bottle. Three remarks. The law holds while the layer is laminar, $\mathrm{Ra}_H < 10^9$, that is $\Delta T_i H^3 < 0.11\,\mathrm{m^3K}$: always, here. Sugar makes a drink about a third more viscous than water, which lowers K_i by under 10 %, since it enters as a quarter power. And for a can lying on its side, as in the rig, the height of the wall is its diameter, and the constant changes a little (Churchill & Chu give a separate law for horizontal cylinders).

5.2 When the drink's film dominates. If R_i is much larger than $R_c + R_w + R_o$, almost all the temperature difference falls across the drink's film, $\Delta T_i \approx \theta$. Then $\dot{Q}_{db} = h_i A_i \theta = K_i A_i\,\theta^{1/4}\theta$:

$$\dot{Q}_{db} = K_i A_i\,\theta^{5/4} \equiv a\,\theta^{5/4}.$$

The heat flow grows faster than the difference, because a larger difference also thins the film.

5.3 Integrate with T_b fixed. Result 1 becomes $C_d\,\mathrm{d}\theta/\mathrm{d}t = -a\theta^{5/4}$. Separate and integrate as in Step 4, using $\int \theta^{-5/4}\mathrm{d}\theta = -4\theta^{-1/4}$:

$$\int_{\theta_0}^{\theta} \theta'^{-5/4}\mathrm{d}\theta' = -\frac{a}{C_d}\int_0^t \mathrm{d}t' \;\Rightarrow\; \theta^{-1/4} = \theta_0^{-1/4} + \frac{at}{4C_d} \;\Rightarrow\; \theta(t) = \theta_0\left(1 + \frac{t}{t^*}\right)^{-4}.$$

The last step factored out $\theta_0^{-1/4}$ and raised both sides to the power -4, which defines

$$t^* = \frac{4C_d}{a\,\theta_0^{1/4}} = 4\tau_0, \qquad \tau_0 = \frac{C_d}{h_i(\theta_0)A_i}.$$

τ_0 is the time constant the drink would have if its film kept its starting coefficient. The curve starts like an exponential with that time constant and then slows, as the film thickens.

5.4 Any exponent. For $\dot{Q} = a\theta^n$ with $n \neq 1$, the same steps with $\int \theta^{-n}\mathrm{d}\theta = \theta^{1-n}/(1-n)$ give

$$\theta^{1-n} = \theta_0^{1-n} + (n-1)\frac{a}{C_d}t \;\Rightarrow\; \theta(t) = \theta_0\big[1 + (n-1)t/\tau_0\big]^{-1/(n-1)}, \qquad \tau_0 = \frac{C_d}{a\theta_0^{n-1}}.$$

As $n \to 1$, $[1 + (n-1)x]^{-1/(n-1)} \to e^{-x}$ and Result 4 comes back.

5.5 Testing the exponent. Taking logarithms of $C_d\,\mathrm{d}T_d/\mathrm{d}t = -a(T_d - T_b)^n$,

$$\ln\left(-\frac{\mathrm{d}T_d}{\mathrm{d}t}\right) = \ln\frac{a}{C_d} + n\ln(T_d - T_b),$$

so the slope of the log of the cooling rate against the log of the difference is n: 1 for a constant U, $\frac{5}{4}$ for a convection-limited drink.

5.6 Mixed resistances. When the other resistances are not small, the drink's film sees only part of θ: what is left after the drop $\dot{Q}_{db}(R_c + R_w + R_o)$ across the rest. That makes $\dot{Q}_{db}$ implicit,

$$\dot{Q}_{db} = K_i A_i\big[\theta - \dot{Q}_{db}\,(R_c + R_w + R_o)\big]^{5/4}, \qquad 1 \le n_{\text{eff}} \le \tfrac{5}{4},$$

and the effective exponent lies between the two limits.

5.7 Bath side, unstirred. The same reasoning applies to the brine on the outside of the can, except that its density difference $\Delta\rho$ comes from two causes: the brine at the wall is warmer, and the ice it melts dilutes it. Both make it lighter, so it rises along the wall. Writing the Rayleigh number with $\Delta\rho$ directly,

$$\mathrm{Nu}_o = \frac{h_o H}{k_b} = C_o\left(\frac{g\,\Delta\rho\,H^3}{\rho_b\nu_b\alpha_b}\right)^{1/4}$$

$$\Rightarrow\; h_o = C_o k_b\left(\frac{g}{\rho_b\nu_b\alpha_b}\right)^{1/4}\Delta\rho^{1/4}H^{-1/4} = 118\,\Delta\rho^{1/4}H^{-1/4}$$

in $\mathrm{W\,m^{-2}K^{-1}}$, with $\Delta\rho$ in $\mathrm{kg\,m^{-3}}$ and H in m, using eutectic brine's properties ($C_o = 0.64$, $k_b = 0.54$, $\rho_b = 1182$, $\nu_b = 4.6 \times 10^{-6}$, $\alpha_b = 1.38 \times 10^{-7}$). $\Delta\rho$ stays a symbol: it depends on how warm the wall is and how much meltwater it makes. On a can 11.7 cm tall, a few $\mathrm{kg\,m^{-3}}$ gives h_o of 250–300 $\mathrm{W\,m^{-2}K^{-1}}$.

5.8 Caveat: β changes with temperature. Water is densest at 4 °C, where $\beta = 0$; below that it expands on cooling. Dissolved sugar and CO_2 move the maximum a little lower. As the drink approaches it the buoyancy in its film vanishes, and $\beta\,\Delta T_i$ must be replaced by the density difference itself:

$$\mathrm{Ra}_H = \frac{g\,[\rho_d(T_{wi}) - \rho_d(T_d)]\,H^3}{\rho_d\,\nu\,\alpha}.$$

Near the maximum the film coefficient falls sharply, so fit n only over the early part of a run.

Result 5 — convection closure and fixed-bath power law

$$h_i = K_i (T_d - T_{wi})^{1/4}, \qquad K_i = 112\, H^{-1/4}$$

$$\dot{Q}_{db} = a\theta^n,\ 1 \le n \le \tfrac{5}{4}, \qquad \theta(t) = \theta_0 \left(1 + \frac{t}{t^*}\right)^{-4} \quad (n = \tfrac{5}{4})$$

What it means. With nothing stirred, the films set the rate, and each grows as the quarter power of its own temperature drop. A constant U (Result 2) is the simplest fit to a run; the power law is the physically expected shape, and the exponent test of 5.5 tells them apart.

Step 6 — Phase equilibrium of the bath

The question. At what temperature can ice and brine of a given salt content coexist, how much salt can brine hold, and what is the coldest the bath can get?

6.1 Equilibrium condition. Water molecules can pass between the ice and the brine. They stop moving on balance when water has the same *chemical potential* — Gibbs energy per mole — in both. Ice is pure, so its chemical potential is that of pure ice, μ_{ice}. In brine the dissolved ions dilute the water and lower its chemical potential below that of pure liquid water, μ_w^*, by $R\mathcal{T}\ln a_w$, where a_w is the water's activity ($a_w = 1$ for pure water, less in brine) and $R = 8.314\,\text{J mol}^{-1}\text{K}^{-1}$. Equilibrium at absolute temperature $\mathcal{T}$ needs

$$\mu_{\text{ice}}(\mathcal{T}) = \mu_w^*(\mathcal{T}) + R\mathcal{T}\ln a_w \ \Rightarrow\ R\mathcal{T}\ln a_w = -\Delta g_{\text{fus}}(\mathcal{T}),$$

where $\Delta g_{\text{fus}} = \mu_w^* - \mu_{\text{ice}}$ is the Gibbs energy of melting one mole of ice.

6.2 How Δg_{fus} changes with temperature. The Gibbs–Helmholtz relation, $\mathrm{d}(\Delta g/\mathcal{T})/\mathrm{d}\mathcal{T} = -\Delta h/\mathcal{T}^2$, links it to the molar heat of melting, $\Delta h_{\text{fus}} = L_f M_w = 6009\,\text{J mol}^{-1}$. Pure ice and water coexist at $\mathcal{T}_0 = 273.15\,\text{K}$, so $\Delta g_{\text{fus}}(\mathcal{T}_0) = 0$. Integrating from $\mathcal{T}_0$ with Δh_{fus} constant, and using 6.1,

$$\frac{\Delta g_{\text{fus}}}{\mathcal{T}} = \Delta h_{\text{fus}}\left(\frac{1}{\mathcal{T}} - \frac{1}{\mathcal{T}_0}\right) \ \Rightarrow\ \ln a_w = -\frac{\Delta h_{\text{fus}}}{R}\left(\frac{1}{\mathcal{T}} - \frac{1}{\mathcal{T}_0}\right),$$

$$\ln a_w = -722.7\left(\frac{1}{\mathcal{T}} - \frac{1}{273.15}\right)$$

with $\mathcal{T}$ in K. Brine of lower water activity is in equilibrium with ice at a lower temperature. Strictly Δh_{fus} is not constant: it falls by $(c_w - c_i)M_w = 39\,\text{J mol}^{-1}$ per kelvin below 0 °C, about 14 % by the eutectic — one reason the measured curve of 6.4 is used in the model.

6.3 The dilute limit: Blagden's law. In dilute brine, a_w is close to the mole fraction of water, $1 - x$, where x is the mole fraction of dissolved particles, and $\ln(1-x) \approx -x$. Per kg of water there are $\nu_i b$ moles of particles, b being the molality (moles of NaCl per kg of water), against $1/M_w$ moles of water, so $x \approx \nu_i b M_w$. Also $1/\mathcal{T} - 1/\mathcal{T}_0 = (\mathcal{T}_0 - \mathcal{T})/(\mathcal{T}\mathcal{T}_0) \approx \Delta T_f/\mathcal{T}_0^2$ for a small drop ΔT_f. Equating,

$$\Delta T_f = \underbrace{\frac{R\mathcal{T}_0^2 M_w}{\Delta h_{\text{fus}}}}_{K_f} \nu_i b,$$

$$K_f = \frac{8.314 \times 273.15^2 \times 0.018015}{6009} = 1.86\ \text{K kg mol}^{-1}.$$

The molality of brine of salt fraction w is $b = (w/M_s)/(1-w) = 17.11\,w/(1-w)\,\text{mol kg}^{-1}$: per kg of brine, w kg of salt is w/M_s moles, dissolved in $1 - w$ kg of water. With $\nu_i = 2$,

$$\Delta T_f = 2 \times 1.86 \times 17.11\,\frac{w}{1-w} = 63.6\,\frac{w}{1-w}\ \text{K}.$$

At 1 wt % this gives 0.64 K; the measured drop is 0.59 K, because the ions are not independent particles. The osmotic coefficient φ (≈ 0.93 here) corrects for that, $\ln a_w = -\varphi \nu_i b M_w$ exactly, and then 6.2 gives, for real brine,

$$\frac{1}{\mathcal{T}_{\mathrm{liq}}} = \frac{1}{\mathcal{T}_0} + \frac{R}{\Delta h_{\mathrm{fus}}} \varphi\, \nu_i\, b\, M_w,$$

$$\frac{1}{\mathcal{T}_{\mathrm{liq}}} = \frac{1}{273.15} + 8.53 \times 10^{-4}\, \varphi(w)\, \frac{w}{1-w} \quad \mathrm{K}^{-1}.$$

6.4 The measured freezing curve (Bodnar 1993). Rather than tabulate φ, the model uses a fit to measured freezing points. With $\vartheta = -T_{\mathrm{liq}}$ the number of degrees below zero and $W = 100w$,

$$W = 1.78\,\vartheta - 0.0442\,\vartheta^2 + 5.572 \times 10^{-4}\,\vartheta^3.$$

Given w, the cubic is solved for ϑ between 0 and 21.2. Its slope follows from $\mathrm{d}W/\mathrm{d}\vartheta = 1.78 - 0.0884\,\vartheta + 1.672 \times 10^{-3}\,\vartheta^2$, with $\mathrm{d}W = 100\,\mathrm{d}w$ and $\mathrm{d}T_{\mathrm{liq}} = -\mathrm{d}\vartheta$:

$$\frac{\mathrm{d}T_{\mathrm{liq}}}{\mathrm{d}w} = -\frac{100}{1.78 - 0.0884\,\vartheta + 1.672 \times 10^{-3}\,\vartheta^2} \quad \mathrm{K}.$$

It is negative — more salt, colder freezing point — and steepens from 0.56 K per wt % near 0 °C to 1.52 K per wt % near the eutectic.

6.5 The phase rule. Gibbs's phase rule counts how many quantities can still be chosen freely, F, for C components in P phases: $F = C - P + 2$. The components are water and NaCl, $C = 2$, and the pressure is fixed, which uses up one: $F = 3 - P$.

Phases present	P	F	What is fixed
brine	1	2	nothing: T and w are both free
ice + brine	2	1	the temperature by the salt: $T = T_{\mathrm{liq}}(w)$
brine + solid salt	2	1	the salt by the temperature: $w = w_{\mathrm{sat}}(T)$
ice + brine + solid salt	3	0	everything: $T = T_E$, $w = w_E$
ice + solid salt	2	1	T free, below T_E

With ice, brine and salt all present, nothing is left to choose: the temperature is pinned at the eutectic, whatever heat comes in, until one of the three phases runs out.

6.6 Eutectic and solubility. The eutectic is where the freezing curve meets the solubility curve:

$$T_{\mathrm{liq}}(w_E) = T_E, \qquad w_{\mathrm{sat}}(T_E) = w_E, \qquad T_E = -21.2\,^{\circ}\mathrm{C}, \quad w_E = 0.2318.$$

Between the eutectic and the peritectic (+0.1 °C, $w_p = 0.26285$) the solubility is taken as the straight line between them, for $-21.2 \le T \le 0.1$ °C:

$$w_{\mathrm{sat}}(T) \approx w_E + (w_p - w_E)\frac{T - T_E}{T_p - T_E} = 0.2318 + 1.458 \times 10^{-3}\,(T + 21.2).$$

Saturated brine is a little saltier when warmer: 0.0015 per kelvin. At the peritectic it matches the handbook solubility of halite at 0 °C, 35.65 g per 100 g of water, $w = 0.2628$.

Result 6 — phase constraints

ice + brine: $T_b = T_{\mathrm{liq}}(w)$; brine + salt: $w = 0.2318 + 1.458 \times 10^{-3}(T_b + 21.2)$;
all three: $T_b = -21.2\,^{\circ}\mathrm{C}$, $w = 0.2318$

What it means. The phase diagram ties the temperature to the salt content whenever two phases coexist, and pins both at the eutectic when three do. That is what makes a salted ice bath a thermostat.

Step 7 — Bath mass balances

The question. How are the amounts of ice, brine and salt related, given that water and salt are neither created nor destroyed?

Water is in the ice and in the brine; salt is in the brine and in the undissolved crystals. A kilogram of brine holds $1 - w$ kg of water and w kg of salt. Counting each substance,

$$m_W = m_i + (1-w)m_l, \qquad m_S = w\,m_l + m_h, \qquad m_W + m_S = m_i + m_l + m_h,$$

the third being the sum of the first two. While no salt is undissolved ($m_h = 0$), these give the brine mass and its salt fraction from the ice alone:

$$m_l = m_W + m_S - m_i, \qquad w = \frac{m_S}{m_W + m_S - m_i}.$$

Melting a mass $\mathrm{d}m_m$ of ice ($\mathrm{d}m_i = -\mathrm{d}m_m$) adds as much water to the brine, and the same salt is spread through more brine. Differentiating,

$$\frac{\mathrm{d}m_l}{\mathrm{d}m_m} = 1, \qquad \frac{\mathrm{d}w}{\mathrm{d}m_m} = -\frac{m_S}{m_l^2} = -\frac{w}{m_l}.$$

Melting always dilutes the brine. Whether the overall salt fraction $\bar{w} = m_S/(m_W + m_S)$ lies below or above w_E decides which phase runs out first on the eutectic plateau (Step 9).

Result 7 — mass constraints

$$m_W = m_i + (1-w)m_l, \quad m_S = w\,m_l + m_h, \quad 0 \le m_i \le m_W, \quad 0 \le m_h \le m_S$$

What it means. Two numbers, the water and the salt put in, fix the bath's make-up; the state only decides how they are shared out between ice, brine and crystals.

Step 8 — Bath enthalpy and the bath energy equation

The question. How much heat does the bath hold, and how does it change?

8.1 Specific enthalpies. Each is counted from the reference (liquid water and solid salt at 0 °C), in J kg^{-1}, with T in °C:

$$h_{\mathrm{ice}} = 2000\,T - 333\,550, \qquad h_{\mathrm{salt}} = 864\,T, \qquad h_{\mathrm{brine}} = (4180 - 4400\,w)\,T + 66\,400\,w.$$

Ice at 0 °C holds less heat than water at 0 °C by the latent heat L_f, and cooling it removes $c_i = 2000$ per kelvin. Salt starts at the reference and has only its sensible heat, $c_h = 864$ per kelvin. Brine is made from water and salt at the reference: dissolving its w kg of salt absorbs $\Delta h_s = 66\,400$ per kg of salt, and it then warms with its own specific heat $c_b(w)$. The heat of diluting brine further is left out.

The straight line $c_b(w) = (1-w)\,c_w + w\,(c_w - 4400)$ can be read as water at $c_w = 4180$ plus dissolved salt at $c_w - 4400 = -220$ J kg^{-1}K^{-1}. The negative number is not a heat capacity of salt: it is how much dissolved ions lower the heat capacity of the water round them. It has one useful consequence: at fixed salt, $m_l c_b(w) = 4180\,m_l - 4400\,m_S$, so each kilogram of water added to the brine adds exactly c_w to its heat capacity.

8.2 Total enthalpy. Enthalpy is extensive: the bath's is the sum of mass times specific enthalpy over its parts, all at the one temperature T_b (A3):

$$\begin{aligned}\mathcal{H} = {} & m_i(2000\,T_b - 333\,550) + m_l\left[(4180 - 4400\,w)\,T_b + 66\,400\,w\right] \\ & + 864\,m_h T_b \quad [+\,C_t T_b].\end{aligned}$$

The last term is there if the tank glass is counted with the bath (A2).

8.3 First law for the bath. The bath is at constant pressure and does no work: as in Step 1, its enthalpy rises by the heat that enters it, from the drink and from the room. There is no stirrer, so no other energy enters.

$$\frac{\mathrm{d}\mathcal{H}}{\mathrm{d}t} = \dot{Q}_{db} + \dot{Q}_{\mathrm{leak}}.$$

8.4 The leak through the box. The box is a resistance between the room and the bath (A4), with conductance UA_{box} (Step 12):

$$\dot{Q}_{\mathrm{leak}} = UA_{\mathrm{box}}(T_{\mathrm{env}} - T_b).$$

8.5 Energy check. Adding Result 1, $C_d\,\mathrm{d}T_d/\mathrm{d}t = -\dot{Q}_{db}$, the drink's term cancels: $\mathrm{d}(C_dT_d + \mathcal{H})/\mathrm{d}t = \dot{Q}_{\mathrm{leak}}$. If the box were perfect ($UA_{\mathrm{box}} = 0$), $C_dT_d + \mathcal{H}$ would stay constant — what the drink loses, the bath gains. Any computation must pass this check.

Result 8 — bath energy equation

$$\frac{\mathrm{d}\mathcal{H}}{\mathrm{d}t} = UA(T_d - T_b) + UA_{\mathrm{box}}(T_{\mathrm{env}} - T_b)$$

$$\mathcal{H} = m_i(2000\,T_b - 333\,550) + m_l\big[(4180 - 4400\,w)\,T_b + 66\,400\,w\big] + 864\,m_hT_b$$

What it means. The bath's state variable is the heat it holds, not its temperature. Heat in raises $\mathcal{H}$; what that does to T_b depends on whether it warms the bath or melts ice, which Step 9 decides.

Step 9 — Inverting $\mathcal{H}$: the regime map

The question. Given the bath's enthalpy and what was put in, what is its temperature, and how much ice, brine and salt does it hold?

There are five unknowns, T_b, m_i, m_l, m_h and w. The two mass balances (Step 7) and the enthalpy (Step 8) give three equations; the phase rule (Step 6) supplies the other two from whichever phases are present (a missing phase supplies its own zero mass). So the state is fixed:

$$(T_b,\, m_i,\, m_l,\, m_h,\, w) = \mathcal{S}(\mathcal{H};\, m_W,\, m_S).$$

Which phases are present depends on $\mathcal{H}$; the five possibilities are the regimes below, from warm to cold.

R-I — liquid only. No ice, no crystals: $m_i = m_h = 0$, $m_l = m_W + m_S$, $w = \bar{w}$. The enthalpy is that of the brine alone, which is linear in T_b:

$$T_b = \frac{\mathcal{H} - 66\,400\,m_S}{(m_W + m_S)(4180 - 4400\,\bar{w})}.$$

R-II — ice and brine on the freezing curve. No crystals ($m_h = 0$); this regime exists only if $\bar{w} < w_E$. The brine sits on the freezing curve, $T_b = T_{\mathrm{liq}}(w)$, and Step 7 gives $m_l = m_S/w$ and $m_i = m_W + m_S - m_S/w$. Putting these into $\mathcal{H}$ makes it a function of w alone:

$$\mathcal{H}_{\mathrm{II}}(w) = \Big(m_W + m_S - \frac{m_S}{w}\Big)(2000\,T_{\mathrm{liq}}(w) - 333\,550) + \frac{m_S}{w}(4180 - 4400\,w)\,T_{\mathrm{liq}}(w) + 66\,400\,m_S.$$

It is solved for w between $\bar{w}$ and w_E. There is exactly one root: adding heat melts ice ($\mathrm{d}\mathcal{H}/\mathrm{d}m_m = L_{\mathrm{eff}} > 0$, Step 10), and melting dilutes the brine ($\mathrm{d}w/\mathrm{d}m_m < 0$), so $\mathcal{H}_{\mathrm{II}}$ falls steadily as w rises. Then $T_b = T_{\mathrm{liq}}(w)$.

R-III — the eutectic plateau. Ice, brine and crystals together: $T_b = -21.2\,^{\circ}\mathrm{C}$ and $w = 0.2318$ (Step 6). Only the brine mass is free, and $\mathcal{H}$ is linear in it. With no brine at all, everything is ice and salt at T_E:

$$\mathcal{H}_{E0} = m_W(c_i T_E - L_f) + m_S c_h T_E = -375\,950\,m_W - 18\,317\,m_S.$$

Each kilogram of eutectic brine is made from $1 - w_E$ kg of ice and w_E kg of salt at T_E; the enthalpy this takes is the brine's minus theirs, $\Lambda_E = h_{\mathrm{brine}} - (1 - w_E)h_{\mathrm{ice}} - w_E h_{\mathrm{salt}}$:

$$\Lambda_E = (1 - w_E)L_f + w_E\Delta h_s + T_E\big[c_b(w_E) - (1 - w_E)c_i - w_E c_h\big].$$

The first term melts 0.768 kg of ice, 256 230 J; the second dissolves 0.232 kg of salt, 15 390 J; the third, $-21.2 \times (3160 - 1536 - 200) = -30\,180$ J, is negative because the brine has a larger heat capacity than the ice and salt it came from, and at T_E, below the reference, a larger heat capacity means less enthalpy. In all, $\Lambda_E = 241\,450\,\mathrm{J\,kg^{-1}}$, and

$$m_l = \frac{\mathcal{H} - \mathcal{H}_{E0}}{241\,450}, \qquad m_i = m_W - 0.7682\,m_l, \qquad m_h = m_S - 0.2318\,m_l.$$

Heat moves the bath along the plateau by making or freezing eutectic brine. If $\bar{w} < w_E$ the salt runs out first, and the bath leaves into R-II; if $\bar{w} > w_E$ the ice runs out first, and it leaves into R-IV.

R-IV — saturated brine and crystals, no ice. Only if $\bar{w} > w_E$. All the water is liquid, $m_l = m_W/(1 - w)$, and the brine is saturated, $w = w_{\mathrm{sat}}(T_b) = 0.2318 + 1.458 \times 10^{-3}(T_b + 21.2)$; the rest of the salt is crystals, $m_h = m_S - w\,m_l$. Then $\mathcal{H}$ is a function of T_b alone:

$$\begin{aligned}\mathcal{H}_{\mathrm{IV}}(T_b) = {} & \frac{m_W}{1 - w}\big[(4180 - 4400\,w)\,T_b + 66\,400\,w\big] \\ & + \Big(m_S - \frac{w\,m_W}{1 - w}\Big)864\,T_b,\end{aligned}$$

solved for T_b between T_E and $T_\star$, the temperature at which all the salt would just dissolve, $w_{\mathrm{sat}}(T_\star) = \bar{w}$, that is $T_\star = -21.2 + (\bar{w} - 0.2318)/(1.458 \times 10^{-3})$.

R-V — fully solid. No brine ($m_l = 0$): all the water is ice and all the salt crystals, below T_E. The enthalpy is linear in T_b:

$$T_b = \frac{\mathcal{H} + 333\,550\,m_W}{2000\,m_W + 864\,m_S}.$$

Regime boundaries. Each threshold is the enthalpy at which a phase appears or disappears, found by putting that state into the formula of either neighbouring regime; the state is continuous across every boundary.

Bath type	Regime	Range	Threshold
$\bar{w} < w_E$	R-I	$\mathcal{H} \geq \mathcal{H}_A$	$\mathcal{H}_A = (m_W + m_S)(4180 - 4400\,\bar{w})T_{\mathrm{liq}}(\bar{w}) + 66\,400\,m_S$: first ice
	R-II	$\mathcal{H}_B \leq \mathcal{H} < \mathcal{H}_A$	$\mathcal{H}_B = \mathcal{H}_{E0} + 1\,041\,600\,m_S$: all salt in eutectic brine
	R-III	$\mathcal{H}_{E0} \leq \mathcal{H} < \mathcal{H}_B$	
	R-V	$\mathcal{H} < \mathcal{H}_{E0}$	
$\bar{w} > w_E$	R-I	$\mathcal{H} \geq \mathcal{H}_C$	$\mathcal{H}_C = (m_W + m_S)(4180 - 4400\,\bar{w})T_\star + 66\,400\,m_S$: first crystal
	R-IV	$\mathcal{H}_D \leq \mathcal{H} < \mathcal{H}_C$	$\mathcal{H}_D = \mathcal{H}_{E0} + 314\,300\,m_W$: all water in eutectic brine
	R-III	$\mathcal{H}_{E0} \leq \mathcal{H} < \mathcal{H}_D$	
	R-V	$\mathcal{H} < \mathcal{H}_{E0}$	

The numbers are $\Lambda_E/w_E = 1\,041\,600$ and $\Lambda_E/(1 - w_E) = 314\,300\,\mathrm{J\,kg^{-1}}$: the enthalpy of the brine that holds all the salt, or all the water, at the eutectic.

Result 9 — bath closure

$$T_b = T_b(\mathcal{H}), \quad m_i = m_i(\mathcal{H}), \quad m_h = m_h(\mathcal{H}), \quad w = w(\mathcal{H}) \quad \text{via R-I to R-V}$$

What it means. The enthalpy alone, with the two masses, tells everything about the bath. This is why the model steps $\mathcal{H}$ forward in time rather than T_b: $\mathcal{H}$ always changes smoothly, while T_b stops dead on the plateau.

Step 10 — Effective latent heat, bath drift and the blank run

The question. On the freezing curve, how much heat melts a kilogram of ice, and how fast does the bath warm?

10.1 Differentiate along the freezing curve. In R-II all the salt is dissolved, so $m_l w = m_S$ is fixed, and by 8.1 $m_l(4180 - 4400w) = 4180\,m_l - 4400\,m_S$. The enthalpy is $\mathcal{H} = m_i(c_i T_b - L_f) + (4180\,m_l - 4400\,m_S)T_b + 66\,400\,m_S$, with $m_l = m_W + m_S - m_i$. A small change of ice and temperature changes it by

$$\mathrm{d}\mathcal{H} = [h_{\mathrm{ice}}(T_b) - c_w T_b]\,\mathrm{d}m_i + \underbrace{(m_i c_i + m_l c_b(w))}_{C_B}\,\mathrm{d}T_b.$$

The first term: a kilogram of ice that melts becomes a kilogram of water in the brine, at heat content $c_w T_b$ in place of $h_{\mathrm{ice}}(T_b)$. The second warms the ice and brine already there; C_B is their heat capacity.

10.2 Heat per kilogram melted. With $\mathrm{d}m_m = -\mathrm{d}m_i$ and $h_{\mathrm{ice}} = c_i T_b - L_f$:

$$L_{\mathrm{eff}} \equiv \frac{\mathrm{d}\mathcal{H}}{\mathrm{d}m_m} = L_f + (c_w - c_i)T_b + C_B \frac{\mathrm{d}T_b}{\mathrm{d}m_m},$$
$$L_{\mathrm{eff}} = 333\,550 + 2180\,T_b + C_B \frac{\mathrm{d}T_b}{\mathrm{d}m_m}.$$

Three parts: the latent heat at 0 °C; a correction because the ice melts at T_b, not 0 °C (below zero the latent heat is smaller by $c_w - c_i = 2180\,\mathrm{J\,kg^{-1}}$ per kelvin, Kirchhoff's law, so this term is negative); and the heat that warms the whole bath, because melting dilutes the brine and so raises its freezing point. The last part makes L_{eff} larger than L_f: for the 20 Oct rig-note charge at −17.8 °C, $L_{\mathrm{eff}} = 411\,500\,\mathrm{J\,kg^{-1}}$, 23 % more.

10.3 Warming per kilogram melted. By the chain rule, with $\mathrm{d}w/\mathrm{d}m_m = -w/m_l$ (Step 7) and the slope of the freezing curve (6.4),

$$\frac{\mathrm{d}T_b}{\mathrm{d}m_m} = \frac{\mathrm{d}T_{\mathrm{liq}}}{\mathrm{d}w}\,\frac{\mathrm{d}w}{\mathrm{d}m_m} = \frac{100\,w}{m_l\,(1.78 - 0.0884\,\vartheta + 1.672 \times 10^{-3}\,\vartheta^2)} > 0.$$

Melting always warms a bath on the freezing curve, and more so the less brine there is.

10.4 Effective heat capacity in every regime. The heat that raises the bath's temperature by one kelvin, $C_{\mathrm{eff}} = \mathrm{d}\mathcal{H}/\mathrm{d}T_b$, follows from differentiating each regime's enthalpy:

$$C_{\mathrm{eff}} = \begin{cases} (m_W + m_S)(4180 - 4400\,\bar{w}) & \text{R-I} \\ C_B + \dfrac{333\,550 + 2180\,T_b}{\mathrm{d}T_b/\mathrm{d}m_m} & \text{R-II} \\ \infty & \text{R-III} \\ m_l c_b(w) + 864\,m_h + \dfrac{1.458 \times 10^{-3}\,m_W}{(1-w)^2}\,(66\,400 - 1084\,T_b) & \text{R-IV} \\ 2000\,m_W + 864\,m_S & \text{R-V} \end{cases}$$

R-I and R-V are plain heat capacities. In R-II, warming by one kelvin needs $1/(\mathrm{d}T_b/\mathrm{d}m_m)$ kg of ice melted, each costing $L_f + (c_w - c_i)T_b$, on top of C_B: this is usually much larger than C_B. On the plateau no amount of heat changes the temperature. In R-IV, warming dissolves more salt, $\mathrm{d}(w\,m_l)/\mathrm{d}T_b = m_W w'_{\mathrm{sat}}/(1-w)^2$ with $w'_{\mathrm{sat}} = 1.458 \times 10^{-3}\,\mathrm{K}^{-1}$, each kilogram absorbing Δh_s and changing the sensible heat by $(-220 - 864)T_b = -1084\,T_b$ (dissolved instead of solid salt, 8.1).

10.5 The plateau as a thermostat. On the plateau the heat coming in, $\dot{Q}_{\mathrm{in}} = \dot{Q}_{db} + \dot{Q}_{\mathrm{leak}}$, does not warm the bath; it makes eutectic brine at Λ_E per kilogram, using ice and salt in the ratio $1 - w_E$ to w_E:

$$\frac{\mathrm{d}m_l}{\mathrm{d}t} = \frac{\dot{Q}_{\mathrm{in}}}{241\,450}, \qquad \frac{\mathrm{d}m_i}{\mathrm{d}t} = -\frac{\dot{Q}_{\mathrm{in}}}{314\,300}, \qquad \frac{\mathrm{d}m_h}{\mathrm{d}t} = -\frac{\dot{Q}_{\mathrm{in}}}{1\,041\,600} \quad (\mathrm{kg\,s^{-1}}).$$

The undissolved salt $m_h(0)$ is the buffer: it lasts

$$t_{\mathrm{buffer}} = \frac{m_h(0)\,\Lambda_E}{w_E\,\bar{\dot{Q}}_{\mathrm{in}}} = \frac{1.042 \times 10^6\, m_h(0)}{\bar{\dot{Q}}_{\mathrm{in}}} \ \mathrm{s},$$

with $m_h(0)$ in kg and the mean heat flow in W. Each 100 g of spare salt holds the bath at $-21.2\,^\circ\mathrm{C}$ against 1 W for 29 hours.

10.6 The blank run. With no drink in the bath and no stirrer, only the leak brings heat in. On the freezing curve, $\dot{Q}_{\mathrm{leak}} = L_{\mathrm{eff}}\,\mathrm{d}m_m/\mathrm{d}t$, and since $\mathrm{d}m_m/\mathrm{d}t = \dot{T}_b/(\mathrm{d}T_b/\mathrm{d}m_m)$,

$$\dot{Q}_{\mathrm{leak}} = L_{\mathrm{eff}}\frac{\mathrm{d}T_b/\mathrm{d}t}{\mathrm{d}T_b/\mathrm{d}m_m} = C_{\mathrm{eff}}\frac{\mathrm{d}T_b}{\mathrm{d}t} \ \Rightarrow\ UA_{\mathrm{box}} = \frac{C_{\mathrm{eff}}\,\dot{T}_b}{T_{\mathrm{env}} - T_b}.$$

A logged warming rate $\dot{T}_b$ therefore measures the box. Using L_f in place of L_{eff} would miss the heat that warms the whole bath and underestimate the leak.

Result 10 — explicit bath temperature equation

$$C_{\mathrm{eff}}(T_b)\frac{\mathrm{d}T_b}{\mathrm{d}t} = UA(T_d - T_b) + UA_{\mathrm{box}}(T_{\mathrm{env}} - T_b) \qquad (\dot{T}_b = 0 \text{ in R-III})$$

What it means. The bath behaves like a body of heat capacity C_{eff}, but one that changes with the regime: huge on the freezing curve, infinite on the plateau. That is why a bath with ice in it warms so slowly.

Step 11 — Initial state from premixing

The question. What state does the bath start in, once the ice, water and salt are mixed?

11.1 Enthalpy of the ingredients. Before mixing, each ingredient is at its own temperature: ice at T_{i0}, water at T_{w0}, salt at T_{s0}. Using 8.1 for each,

$$\mathcal{H}_0 = m_{i0}(2000\,T_{i0} - 333\,550) + 4180\,m_{w0}T_{w0} + 864\,m_S T_{s0},$$

and all the water is $m_W = m_{i0} + m_{w0}$.

11.2 Adiabatic mixing. In the insulated box, mixing exchanges heat only among the ingredients, so the enthalpy does not change (Step 8 with no heat in). The starting state is the one Step 9 gives for $\mathcal{H}_0$: $(T_{b0}, m_i, m_l, m_h, w)(0) = \mathcal{S}(\mathcal{H}_0; m_W, m_S)$. Enthalpy is a state function, so this does not depend on the order of mixing: adding the salt at once, in stages, or pre-dissolved reaches the same equilibrium, only at different rates.

11.3 Start on the eutectic. The bath starts on the plateau if $\mathcal{H}_0$ falls in its range (Step 9):

$$\mathcal{H}_{E0} \le \mathcal{H}_0 < \mathcal{H}_{E0} + 1\,041\,600\,m_S \quad \text{if } \bar{w} < w_E,$$
$$\mathcal{H}_{E0} \le \mathcal{H}_0 < \mathcal{H}_{E0} + 314\,300\,m_W \quad \text{if } \bar{w} > w_E.$$

11.4 Keep a salt buffer. To start with at least m_{buf} of salt undissolved, the plateau formula $m_h = m_S - 0.2318\, m_l$ with $m_l = (\mathcal{H}_0 - \mathcal{H}_{E0})/241\,450$ must give $m_h \geq m_{\text{buf}}$:

$$\mathcal{H}_0 \leq \mathcal{H}_{E0} + 1\,041\,600\,(m_S - m_{\text{buf}}).$$

11.5 Minimum ice fraction. Write the ice as a fraction $f = m_{i0}/m_W$ of the water. Then $\mathcal{H}_0 = m_W[f(2000\,T_{i0} - 333\,550) + (1-f)\,4180\,T_{w0}] + 864\,m_S T_{s0}$, which falls as f rises. For $\mathcal{H}_0$ to stay below any bound $\mathcal{H}_{\max}$ (from 11.3 or 11.4),

$$f \geq f_{\text{crit}} = \frac{4180\,m_W T_{w0} + 864\,m_S T_{s0} - \mathcal{H}_{\max}}{m_W\,(4180\,T_{w0} - 2000\,T_{i0} + 333\,550)}.$$

11.6 Melting against dissolving. When the mixture cools itself, the heat comes from melting Δm of ice and dissolving $m_{S,\text{dis}}$ of salt. The share due to melting is

$$\chi_{\text{lat}} = \frac{333\,550\,\Delta m}{333\,550\,\Delta m + 66\,400\,m_{S,\text{dis}}}.$$

Per kilogram, melting ice absorbs five times as much heat as dissolving salt.

Result 11 — initial conditions

$$T_d(0) = T_{d0},$$
$$\mathcal{H}(0) = m_{i0}(2000\,T_{i0} - 333\,550) + 4180\,m_{w0}T_{w0} + 864\,m_S T_{s0}$$

What it means. How cold the bath starts, and whether it starts on the plateau with salt to spare, is set by a single number, the enthalpy of what goes in; a drier, colder charge has less of it.

Step 12 — The box leak

The question. How fast does heat from the room get into the bath?

Heat from the room crosses three layers in series: the film of room air on the outside of the box, the foam, and the air between the foam and the tank. As in Step 2 their resistances add:

$$\frac{1}{U A_{\text{box}}} = \frac{1}{h_{\text{air}} A_{\text{out}}} + \frac{\delta_{\text{box}}}{0.035\,A_{\text{box}}} + \frac{1}{h_{\text{in}} A_{\text{in}}} \approx \frac{\delta_{\text{box}}}{0.035\,A_{\text{box}}},$$

with expanded polystyrene's conductivity, $0.035\,\text{W m}^{-1}\text{K}^{-1}$. The foam dominates: 30 mm of it has $\delta/k = 0.86\,\text{m}^2\text{K W}^{-1}$, against about $1/9 = 0.11$ for room air (free convection plus radiation) and $1/6 = 0.17$ for the air gap inside. Dropping the films overestimates the leak by about a third; gaps round the lid, where heat bypasses the foam, push it the other way.

Two variants for when an assumption is broken. With the lid off (A5), the bath's open surface also takes heat from the air and from water condensing on it: $\dot{Q}_{\text{surf}} = h_s A_s (T_{\text{air}} - T_b) + \dot{m}_v h_{sg}$, with $\dot{m}_v$ the condensation rate and h_{sg} its latent heat. With the drink not fully immersed (A8), its exposed part takes heat from the air: $C_d \dot{T}_d = -UA(T_d - T_b) + UA_{da}(T_{\text{air}} - T_d)$.

Result 12 — the leak

$$\dot{Q}_{\text{leak}} = U A_{\text{box}} (T_{\text{env}} - T_b), \qquad U A_{\text{box}} \approx \frac{0.035\,A_{\text{box}}}{\delta_{\text{box}}}$$

What it means. The leak is proportional to how much colder the bath is than the room, so it is largest exactly when the bath is most useful. It is best measured, with the blank run of 10.6.

Step 13 — The closed model

The question. What is the complete set of equations to integrate?

Two first-order differential equations for the state $(T_d, \mathcal{H})$, closed by one law for the heat flow and one map from $\mathcal{H}$ to the bath's state.

13.1 Governing equations (Results 1 and 8):

$$C_d \frac{\mathrm{d}T_d}{\mathrm{d}t} = -\dot{Q}_{db}, \qquad \frac{\mathrm{d}\mathcal{H}}{\mathrm{d}t} = \dot{Q}_{db} + UA_{\mathrm{box}}(T_{\mathrm{env}} - T_b).$$

13.2 Heat-flow closure — choose one (Results 2 and 5):

$$\text{A (constant conductance):} \quad \dot{Q}_{db} = UA(T_d - T_b), \quad \frac{1}{UA} = R_i + R_c + R_w + R_o$$

$$\text{B (drink-film convection):} \quad \dot{Q}_{db} = 112\, H^{-1/4} A_i \left[(T_d - T_b) - \dot{Q}_{db}(R_c + R_w + R_o)\right]^{5/4}$$

13.3 Bath closure (Results 6, 7 and 9), $(T_b, m_i, m_l, m_h, w) = \mathcal{S}(\mathcal{H}; m_W, m_S)$, in J, kg and °C:

$$\mathcal{S}: \begin{cases} \text{R-I:} & m_i = m_h = 0,\ w = \bar{w},\ T_b = \dfrac{\mathcal{H} - 66\,400\, m_S}{(m_W + m_S)(4180 - 4400\,\bar{w})} \\ \text{R-II:} & m_h = 0,\ m_l = m_S/w,\ T_b = T_{\mathrm{liq}}(w),\ w \text{ from } \mathcal{H} = \mathcal{H}_{\mathrm{II}}(w) \text{ (Step 9)} \\ \text{R-III:} & T_b = -21.2,\ w = 0.2318,\ m_l = \dfrac{\mathcal{H} - \mathcal{H}_{E0}}{241\,450},\ m_i = m_W - 0.7682\, m_l,\ m_h = m_S - 0.2318\, m_l \\ \text{R-IV:} & m_i = 0,\ w = w_{\mathrm{sat}}(T_b),\ T_b \text{ from } \mathcal{H} = \mathcal{H}_{\mathrm{IV}}(T_b) \text{ (Step 9)} \\ \text{R-V:} & m_l = 0,\ T_b = \dfrac{\mathcal{H} + 333\,550\, m_W}{2000\, m_W + 864\, m_S} \end{cases}$$

with the regime chosen by the thresholds of Step 9, and

$$W = 1.78\,\vartheta - 0.0442\,\vartheta^2 + 5.572 \times 10^{-4}\,\vartheta^3 \quad (W = 100w,\ \vartheta = -T_{\mathrm{liq}}),$$

$$w_{\mathrm{sat}}(T) = 0.2318 + 1.458 \times 10^{-3}(T + 21.2),$$

$$\mathcal{H}_{E0} = -375\,950\, m_W - 18\,317\, m_S, \qquad \bar{w} = \frac{m_S}{m_W + m_S}.$$

13.4 Equivalent explicit form. For a hand calculation (Desmos, Mathematica), Result 10 replaces the $\mathcal{H}$ equation:

$$C_d \dot{T}_d = -\dot{Q}_{db}, \qquad C_{\mathrm{eff}}(T_b)\, \dot{T}_b = \dot{Q}_{db} + UA_{\mathrm{box}}(T_{\mathrm{env}} - T_b) \quad (\dot{T}_b = 0 \text{ in R-III}).$$

A program should integrate the $\mathcal{H}$ form, which crosses the plateau without dividing by $C_{\mathrm{eff}} = \infty$.

13.5 Initial conditions (Result 11):

$$T_d(0) = T_{d0}, \qquad m_W = m_{i0} + m_{w0},$$

$$\mathcal{H}(0) = m_{i0}(2000\, T_{i0} - 333\,550) + 4180\, m_{w0} T_{w0} + 864\, m_S T_{s0}.$$

13.6 When to stop, and when the model stops being valid.

- T_d reaches the target T^*: the measured output.
- T_d reaches the drink's freezing point T_{fd} (about $-0.9\,°\mathrm{C}$ for a sugary fizzy drink): beyond it the drink needs its own enthalpy model, like the bath's.
- In a buffered run, $m_h \to 0$: the thermostat is gone and the bath leaves the plateau.

- T_d approaches the density maximum near 4 °C: closure B and assumption A1 weaken (5.8).

13.7 What has to be measured.

Group	Parameters	How they are fixed
Drink	m_d, c_d, m_c	weighing; the drink's specific heat from its label (sugar)
Heat path	UA (A), or A_i, H, R_c, R_w, R_o (B)	fitted on a buffered run, where $T_b = -21.2\,°\mathrm{C}$ is known
Bath	m_W, m_S, m_{i0}, m_{w0}	weighing
Environment	UA_{box}, $T_{\mathrm{env}}(t)$	a blank run (10.6); room temperature logged on channel 4
Start	T_{d0}, T_{i0}, T_{w0}, T_{s0}	logged

Every other constant is in Table 1. Once UA is fixed on a buffered run, an unbuffered run has no free parameter left: it is a prediction.

Result 13 — the model, in one block

$$
\begin{aligned}
C_d\,\dot{T}_d &= -\,UA\,(T_d - T_b), \qquad C_d = m_d c_d + 900\,m_c \\
\dot{\mathcal{H}} &= UA\,(T_d - T_b) + UA_{\mathrm{box}}(T_{\mathrm{env}} - T_b) \\
T_b &= T_b(\mathcal{H};\, m_W,\, m_S) \quad \text{via } \mathcal{S} \\
T_d(0) &= T_{d0}, \qquad \mathcal{H}(0) = \mathcal{H}_0
\end{aligned}
$$

What it means. Five numbers describe the rig (C_d, UA, m_W, m_S, UA_{box}), plus the starting temperatures; the rest is physics that is the same in any laboratory.

Step 14 — Closed-form special cases

The question. When can the equations be solved with a formula, rather than step by step?

14.1 Buffered bath. If the bath stays on the plateau (R-III throughout), $T_b = -21.2\,°\mathrm{C}$ is constant and Result 4 applies directly:

$$T_d(t) = -21.2 + (T_{d0} + 21.2)\,e^{-t/\tau}, \qquad \tau = \frac{C_d}{UA},$$

valid while $t < t_{\mathrm{buffer}}$ (10.5).

14.2 Bath on the freezing curve, sealed box. In R-II over a small range of temperature, C_{eff} is roughly constant; take $UA_{\mathrm{box}} = 0$. Then drink and bath are two heat capacities joined by one conductance. Dividing the two equations of 13.4 by their heat capacities and subtracting,

$$\frac{\mathrm{d}\theta}{\mathrm{d}t} = -UA\Big(\frac{1}{C_d} + \frac{1}{C_{\mathrm{eff}}}\Big)\theta \;\Rightarrow\; \theta = \theta_0 e^{-t/\tau_c}, \qquad \tau_c = \frac{C_d C_{\mathrm{eff}}}{UA\,(C_d + C_{\mathrm{eff}})} = \frac{\tau}{1+\Gamma}, \quad \Gamma = \frac{C_d}{C_{\mathrm{eff}}}.$$

The difference closes faster than with a fixed bath, because the bath warms to meet the drink. Energy conservation (8.5), $C_d T_d + C_{\mathrm{eff}} T_b = \mathrm{const}$, fixes the common final temperature, and each temperature approaches it with the same time constant:

$$T_\infty = \frac{C_d T_{d0} + C_{\mathrm{eff}} T_{b0}}{C_d + C_{\mathrm{eff}}}, \qquad T_d(t) = T_\infty + (T_{d0} - T_\infty)e^{-t/\tau_c}, \qquad T_b(t) = T_\infty - (T_\infty - T_{b0})e^{-t/\tau_c}.$$

The drink reaches T^* only if $T_\infty < T^*$, after $t_{T^*} = \tau_c \ln\left[(T_{d0} - T_\infty)/(T^* - T_\infty)\right]$.

14.3 With the box leak. With T_{env} constant, the two equations of 13.4 are linear in $\mathbf{x} = (T_d, T_b)^{\mathsf{T}}$: $\dot{\mathbf{x}} = \mathsf{A}\mathbf{x} + \mathbf{f}$, with $\mathbf{f} = (0,\ UA_{\text{box}}T_{\text{env}}/C_{\text{eff}})^{\mathsf{T}}$ and

$$\mathsf{A} = \begin{pmatrix} -\dfrac{UA}{C_d} & \dfrac{UA}{C_d} \\[2ex] \dfrac{UA}{C_{\text{eff}}} & -\dfrac{UA + UA_{\text{box}}}{C_{\text{eff}}} \end{pmatrix}.$$

The solution is a steady state plus two decaying exponentials, $e^{\lambda_\pm t}$, where $\lambda_\pm$ are the eigenvalues of A, the roots of $\lambda^2 - (\operatorname{tr}\mathsf{A})\lambda + \det\mathsf{A} = 0$:

$$\lambda_\pm = \tfrac{1}{2}\Big[\operatorname{tr}\mathsf{A} \pm \sqrt{(\operatorname{tr}\mathsf{A})^2 - 4\det\mathsf{A}}\Big],$$
$$\operatorname{tr}\mathsf{A} = -\Big(\frac{UA}{C_d} + \frac{UA + UA_{\text{box}}}{C_{\text{eff}}}\Big), \qquad \det\mathsf{A} = \frac{UA \cdot UA_{\text{box}}}{C_d\, C_{\text{eff}}}.$$

The trace is negative and the determinant positive, so both eigenvalues are real and negative. When the leak is small, the fast one is nearly $\operatorname{tr}\mathsf{A}$: the drink and bath coming together as in 14.2. The slow one is nearly $\det\mathsf{A}/\operatorname{tr}\mathsf{A}$: the room warming both. The steady state is $T_d = T_b = T_{\text{env}}$, reached only after the ice is gone — by which time C_{eff} has changed, so this solution holds only within one regime.

14.4 Fixed bath, convection-limited drink. Result 5: $\theta = \theta_0(1 + t/t^*)^{-4}$, $t^* = 4\tau_0$.

What it means. Each special case isolates one physical effect: the thermostat (14.1), the bath warming to meet the drink (14.2), and the room warming both (14.3). A full run strings them together as the bath moves from regime to regime.